YOUVRAJ SINGH

Software Engineer · backend, data integrity and performance

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SUMMARY

Computer Science graduate who ships systems where the numbers have to be exactly right. Sole developer of three production systems at a Bengaluru food-tech startup: an **append-only stock ledger** that reconciled a month of real sales with an **exact match**, a **payroll engine** whose finalised snapshots are immutable, and a delivery platform, all covered by **1,300+ automated tests**. Grounded in operating systems, database systems and compiler design from my degree; C++ from coursework and algorithm practice; fluent in **Python, TypeScript and SQL**. Looking to work on the engine beneath applications: storage, query execution and performance.

1,300+ automated tests across three production systems	**Exact match** a month of real sales reconciled against the ledger	**₹0.00** payroll difference vs the finance sheet, 50+ staff	**25–30%** load-time and server-response improvements, measured

SKILLS

Languages	**C++** (coursework, data structures & algorithms), **Python**, **TypeScript**, JavaScript, **SQL**, Java
CS fundamentals	Data structures & algorithms, operating systems, database systems (indexing, transactions, ACID, isolation), compiler design, computer architecture
Data & storage	PostgreSQL, MongoDB · schema design, append-only / event-sourced ledgers, immutable snapshots, migrations, audit logs, reconciliation
Performance & testing	Profiling (Chrome DevTools, Lighthouse), latency budgets, pytest, Playwright, Vitest, exact-value end-to-end assertions
Backend & tooling	FastAPI, Node.js, NestJS, REST, JWT & RBAC, rate limiting · Git, GitHub Actions, Docker, AWS (S3), Render, Vercel
Frontend	React, Next.js, Tailwind CSS

EXPERIENCE

Full-Stack Developer Intern · SKOPE Kitchens — Bengaluru · Oct 2026 – Present

- **Designed an append-only stock ledger** (Python/FastAPI, MongoDB) for a multi-brand cloud kitchen: every receipt, issue, transfer, production run, sale and adjustment is a new entry, stock is derived (in − out) and history is never rewritten. **Reconciled a month of real POS sales with an exact match.**
- **Modelled multi-level recipes** (raw material → semi-finished → dish, per-level yield) so a sale consumes ingredients transitively with a shortage check first; a daily close compares physical counts to the ledger and **locks the day**.
- **Built a payroll engine** (TypeScript, PostgreSQL · 22 tables, ~60 endpoints) for 50+ staff: overtime in 30-minute blocks, a GENERATED → PAID state machine, and **finalisation freezes an immutable snapshot**. First live payroll matched the finance sheet **to the paisa**.
- **Wrote a 464-check end-to-end suite** that runs purchase-to-close as each of 6 roles and asserts exact ledger quantities after every step; 1,300+ tests overall (pytest, Playwright, Vitest).
- Production hardening: role checks in API (403) and UI, rate limiting, append-only audit logs, encrypted scheduled backups (GitHub Actions), and a Create React App → Vite migration after the old build broke on a fresh install.

Software Development Intern · HASVIK Technologies — Remote · Jun 2026 – Oct 2026

- Built NestJS REST APIs for a **CRM migration** and **restructured PostgreSQL schemas** for data consistency; every endpoint documented and tested with Swagger.
- **Profiled and refactored** client code, **cutting page load time by 25%**; shipped under senior code review and co-managed AWS deployments.

SELECTED PROJECTS

Instantly · e-commerce platform · Live — *TypeScript, Node.js, MongoDB, Stripe, AWS S3*

- Stripe **signature-verified webhooks** as the only source of truth for "paid"; checkout kept **under 400 ms**.
- Redesigned MongoDB schemas and query paths: **~30% faster server responses**; 25% faster client render with Zustand and TanStack Query.

CarRental · booking platform · Live — *Node.js, Express, MongoDB, JWT*

- 14+ REST endpoints with **date-range overlap validation** (no double bookings) and JWT-scoped access for customers and owners.

Portfolio · 40,000-particle WebGL engine · youvraj-singh.vercel.app — *TypeScript, three.js, GLSL, Next.js*

- One draw call, custom GLSL shaders, a frame-time probe that halves the particle count on slow GPUs, and a render loop that sleeps when idle; 140+ automated tests, Lighthouse 99–100 desktop.

EDUCATION & CERTIFICATIONS

B.Sc. Computer Science · St. Aloysius College, Jabalpur (Rani Durgavati Vishwavidyalaya) — 2020 – 2023

Coursework: data structures & algorithms (C++), operating systems, database management systems, compiler design, computer organisation & architecture.

Workato Foundations · Full Stack Development Bootcamp (Udemy) · Solana Bootcamp (Rise In)